HARSHANA KARUNARATHNE

Software Engineer | Java, Spring Boot, React, Microservices
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PROFESSIONAL SUMMARY

Full-stack Software Engineer with 2+ years of experience designing and delivering enterprise-grade applications with Java, Spring Boot and React. Builds scalable, secure systems end to end, from microservices and REST APIs to event-driven architectures with Apache Kafka, real-time messaging and AI integrations, deployed with Docker on AWS. Experienced in banking and payment systems, multi-tenant platforms, role-based security, audit compliance and modernizing legacy Java applications. Graduate in Statistics and Computer Science.

TECHNICAL SKILLS

Languages: Java (8, 17, 21), TypeScript, JavaScript, Python, SQL

Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Spring AI, Hibernate, REST APIs, Microservices

Frontend: React, Tailwind CSS, TanStack Query, Electron, Thymeleaf

Messaging and Real-Time: Apache Kafka, MQTT, WebSockets, Server-Sent Events (SSE), Webhooks

Databases: MySQL, MSSQL, PostgreSQL, MongoDB, Firebase, Flyway

Cloud and DevOps: AWS, Docker, Kubernetes, CI/CD, Git, GitHub, GitLab

Security and Testing: JWT, Passkeys (WebAuthn), Role-Based Access Control (RBAC), Audit Logging, VAPT Remediation, JUnit, Vitest

AI: Google Gemini, Gemini Live API, LLM Function Calling

WORK EXPERIENCE

Software Engineer, TS Technologies (Pvt) Ltd **Sep 2024 – Present**

Java 21, *Spring Boot*, *Microservices*, *React*, *TypeScript*, *MySQL*, *Electron*

- Full-stack engineer on the company's restaurant commerce product line – a multi-branch POS and an AI voice-ordering kiosk – taking features from design through Spring Boot services, React front ends and Electron hardware integration to release.
- Designed the integration between the POS and kiosk products: versioned REST APIs, webhooks and a shared multi-branch data model, with each product keeping its own database.
- Built reusable React components and Spring Boot modules with Flyway migrations and automated tests (JUnit, Vitest), and contributed to the company's CRM platform.

Software Engineer Intern, Epic Lanka (Pvt) Ltd **Feb 2024 – Sep 2024**

Java 17, *Spring Boot*, *Spring MVC*, *Apache Kafka*, *MSSQL*, *Docker*, *AWS*

- Modernized a centralized terminal management system supporting 5,000+ POS devices by migrating it from Struts (Java 8) to Java 17 with Spring Boot, improving scalability, performance and maintainability.
- Designed and delivered payment processing and billing microservices with Spring Boot and Apache Kafka, automating event-driven transaction workflows.
- Built RESTful APIs and server-rendered views with Spring MVC, Thymeleaf and MSSQL for secure enterprise data handling.
- Enforced enterprise security standards by resolving VAPT findings and implementing audit logging with 100% traceability across terminals.
- Improved the stability of production backend services in a banking environment through performance tuning and debugging.

PROJECTS

TS POS and AI Voice Kiosk – Restaurant Commerce Platform **TS Technologies**

Java 21, *Spring Boot*, *Microservices*, *MySQL*, *Flyway*, *React*, *TypeScript*, *Electron*, *Spring AI*, *Google Gemini Live API*, *SSE*, *MQTT*, *WebSockets*

- Architected the platform as Spring Boot microservices – the POS Core API and the kiosk backend, each with its own database, integrated via a versioned REST and webhook contract – with payment and inventory designed as decoupled services that can be deployed and scaled independently.
- Built the POS suite – sales terminal, Kitchen Display System, admin portal and QR guest menu – with a central pricing engine for taxes, discounts and promotions, and idempotent order and payment APIs that prevent duplicate charges.
- Built AI voice ordering with Spring AI and the Gemini Live API, reducing average order time by 40%, and a recommendation engine that raised average order value by 25%.
- Delivered real-time updates over SSE and MQTT, Electron-driven receipt printers and cash drawers, and multi-branch RBAC with JWT, audit logging and branch-scoped data access.

Customer Relationship Management System **TS Technologies**

- Developed CRM modules for customer management, lead tracking and sales workflow automation.
- Implemented AI-driven lead prioritization to reduce manual lead analysis.
- Optimized database queries and backend services to speed up page loads and reporting, and automated routine business processes.

ATEM Hub – Terminal Management System for BML **Epic Lanka**

- Revamped a centralized platform for managing POS applications and devices, migrating it from Struts (Java 8) to Java 17 with Spring Boot and Spring MVC, with REST APIs, Thymeleaf server-side rendering and MSSQL.

Automatic Bill Settlement **Epic Lanka**

- Built Payment Processor and Billing Mediator microservices for a next-generation payment system using Java 17, Spring Boot, React and Apache Kafka, deployed in Docker containers on AWS with cloud scheduling, load balancing and VAPT-compliant security.

EDUCATION

B.Sc. in Physical Science (Statistics and Computer Science) **2022 – 2025**
University of Kelaniya | Dean's List, awarded every academic year

G.C.E. Advanced Level, Physical Science Stream **2017 – 2020**
Royal College, Colombo | Combined Mathematics A, Chemistry B, Physics C

REFERENCES

Dr. Basitha, Senior Lecturer, University of Kelaniya | basitha@kln.ac.lk
Mr. Muditha Hettiarachchi, Director, Success Delivery Operations, IFS | muditha.hettiarachchi@ifs.com